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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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RESOURCE MANAGEMENT IN A MULTI-USER LINUX-BASED SERVER ENVIRONMENT

CSA0401 - OPERATING SYSTEMS

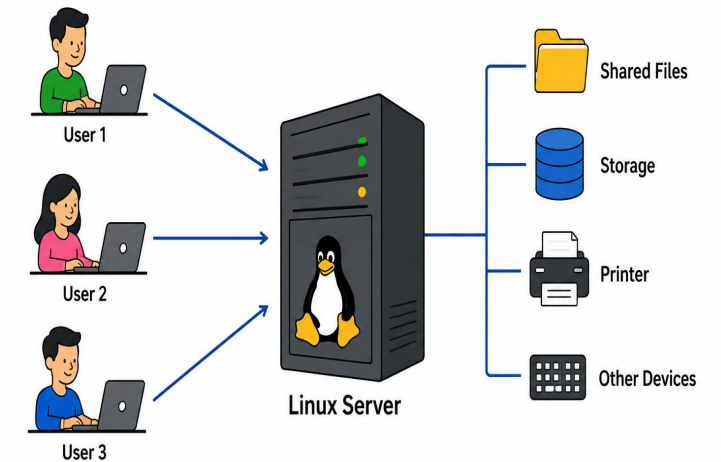
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INTRODUCTION



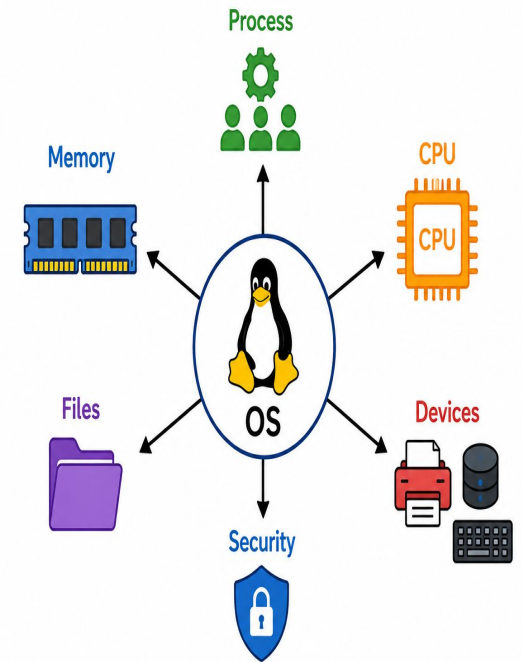
- A **Linux-based server** allows multiple users to access files, applications, and devices at the same time.
- The **Operating System (OS)** manages system resources efficiently and securely.
- It controls the CPU, memory, files, and hardware devices.
- The Linux kernel acts as a bridge between user applications and the hardware.



OS Functions in Resource Management

OS Functions

- The Linux operating system manages system resources by:
 - **Process Management** – Creates and schedules processes.
 - **Memory Management** – Allocates and manages RAM.
 - **CPU Scheduling** – Shares CPU time among users.
 - **File Management** – Controls file storage and permissions.
 - **Device Management** – Manages hardware devices like disks and printers.



FILE AND DEVICE SYSTEM CALLS

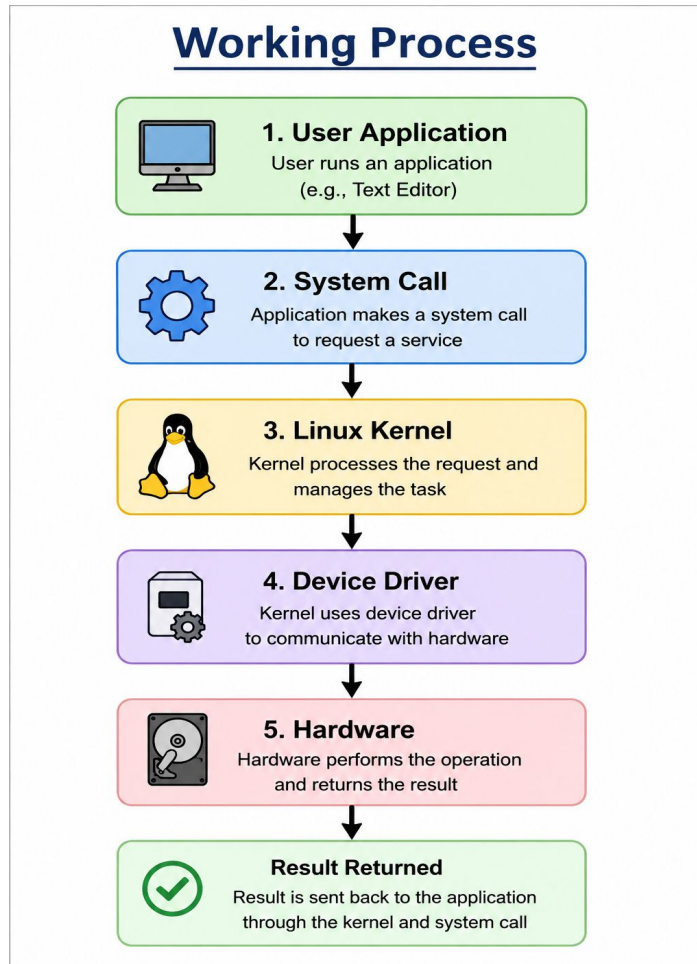
File System Calls

- `open()` – Opens a file.
- `read()` – Reads data from a file.
- `write()` – Writes data to a file.
- `close()` – Closes the file.

Device System Calls

- `open()`
- `read()`
- `write()`
- `ioctl()` – Controls device-specific operations.
- `close()`

Interaction Between User Processes and Kernel



STEPS:

- User runs an application.
- Application makes a system call.
- Kernel processes the request.
- Device driver communicates with hardware.
- Result is returned to the application.

CONCLUSION

- Linux efficiently manages resources in a multi-user environment.
- System calls provide communication between applications and the kernel.
- The kernel ensures secure and efficient access to files and devices.
- This improves system performance, reliability, and security.

